

BUCHANAN FIELD, CA, USA

WMO: 724950

Lat: 37.992N

Lon: 122.055W

Elev: 6

StdP: 101.26

Time Zone: -8.00 (NAP)

Period: 99-19

WBAN: 23254

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.6	(c) 2.0	(d) -7.4	(e) 2.0	(f) 12.7	(g) -5.1	(h) 2.5	(i) 12.3	(j) 8.8	(k) 13.0	(l) 7.7	(m) 11.9	(n) 0.8	(o) 160	(p) 0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.9	(c) 36.6	(d) 19.4	(e) 34.5	(f) 18.8	(g) 32.6	(h) 18.3	(i) 20.5	(j) 33.8	(k) 19.7	(l) 32.2	(m) 19.0	(n) 30.6	(o) 4.5	(p) 300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.7	11.2	23.1	15.0	10.7	22.1	14.4	10.2	21.4	58.9	33.7	56.0	32.2	53.6	30.6	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.9	8.1	7.4	-1.2	40.4	1.2	1.7	-2.0	41.7	-2.7	42.7	-3.4	43.6	-4.2	44.8		
			-2.4	22.0	1.3	1.1	-3.3	22.8	-4.1	23.5	-4.8	24.1	-5.7	25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.4	9.6	11.3	13.5	15.2	18.2	21.3	22.6	22.5	21.5	18.0	12.9	9.5
	DBStd	5.56	2.47	2.55	2.78	3.03	3.03	3.27	2.86	2.48	2.93	2.80	2.91	2.70
	HDD10.0	106	38	14	3	1	0	0	0	0	0	0	8	41
	HDD18.3	1257	272	198	152	104	40	7	2	1	4	41	164	273
	CDD10.0	2426	24	50	112	156	254	339	392	387	346	247	95	26
	CDD18.3	537	0	0	2	9	36	96	135	129	100	30	1	0
	CDH23.3	6216	0	4	41	165	510	1172	1480	1303	1118	403	19	0
	CDH26.7	2877	0	0	3	48	203	575	752	620	542	131	2	0
(14) Wind	WSAvg	3.2	2.0	2.5	2.9	3.5	4.2	4.5	4.6	4.2	3.3	2.5	2.0	2.1
(15) Precipitation	PrecAvg	511	110	91	65	36	19	6	0	1	2	29	59	101
	PrecMax	775	267	328	211	118	75	47	2	21	15	108	173	258
	PrecMin	222	5	0	0	6	0	0	0	0	0	3	0	0
	PrecStd	135	73	66	58	27	21	10	1	4	3	27	42	74
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.1	23.5	26.9	31.0	35.2	38.8	39.2	37.9	37.6	32.8	25.7	18.9
		MCWB	11.0	13.8	16.2	17.6	19.0	20.2	20.3	19.8	19.0	17.2	14.9	12.1
	2%	DB	17.6	20.6	24.4	27.8	31.6	35.6	36.0	35.1	35.1	30.1	23.0	17.2
		MCWB	11.6	13.0	15.1	16.2	17.6	19.1	19.4	19.1	18.3	16.2	14.1	11.8
	5%	DB	16.0	18.4	22.1	24.8	28.7	32.8	33.5	32.6	32.6	27.7	20.7	15.8
		MCWB	11.2	12.1	14.0	15.0	16.5	18.3	19.0	18.5	17.8	15.2	13.1	11.7
	10%	DB	14.5	16.6	19.8	22.1	26.0	29.9	31.1	30.2	29.8	25.4	18.7	14.6
		MCWB	10.7	11.7	13.0	13.9	15.6	17.2	18.4	18.0	17.1	14.8	12.6	11.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.5	15.7	17.4	18.7	20.4	21.5	22.2	20.9	20.9	18.8	16.8	15.3
		MCDB	16.6	20.1	24.5	28.4	32.6	36.3	36.9	34.2	35.2	28.8	22.7	17.0
	2%	WB	13.3	14.5	16.0	16.9	18.6	20.0	20.4	20.0	19.5	17.6	15.7	14.1
		MCDB	15.5	18.2	22.7	26.1	30.2	33.4	33.6	32.4	31.5	26.8	20.3	15.6
	5%	WB	12.3	13.5	14.8	15.7	17.2	18.9	19.5	19.1	18.5	16.7	14.8	13.0
		MCDB	14.3	17.1	20.4	23.6	26.9	31.2	31.9	30.6	29.5	24.9	18.8	14.6
	10%	WB	11.3	12.4	13.7	14.4	16.0	17.8	18.6	18.4	17.7	15.7	13.8	11.9
		MCDB	13.6	15.8	18.5	21.0	24.7	28.8	30.2	29.1	28.1	22.7	17.5	13.6
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.9	10.9	11.7	13.0	14.9	15.9	15.3	15.0	13.5	11.0	9.1
		MCDBR	12.1	13.7	15.1	17.0	18.5	20.0	20.0	19.5	20.1	17.9	14.2	11.0
		MCWBR	7.0	7.5	7.5	7.8	7.2	6.8	6.6	6.3	6.6	6.9	6.9	6.4
	5% WB	MCDBR	8.3	10.6	13.0	15.4	16.9	18.5	19.0	17.6	17.4	14.9	11.2	6.5
		MCWBR	5.7	6.3	7.0	7.5	7.1	6.9	6.7	6.2	6.2	6.3	6.1	4.7
(40) Clear-Sky Solar Irradiance	taub		0.323	0.326	0.340	0.359	0.364	0.361	0.366	0.350	0.348	0.333	0.325	0.326
	taud		2.536	2.529	2.475	2.394	2.407	2.421	2.406	2.444	2.446	2.525	2.530	2.529
	Ebn at Noon		862	907	924	919	916	915	907	917	901	885	852	830
	Edh at Noon		77	87	101	116	117	116	116	109	103	86	77	73
(44) All-Sky Solar Radiation	RadAvg		2.32	3.29	4.67	6.13	7.23	7.98	7.74	6.91	5.74	4.15	2.79	2.05
(45)	RadStd		0.36	0.41	0.45	0.51	0.44	0.33	0.23	0.24	0.20	0.27	0.18	0.31

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (47 neighbors)	+0.37	N/A	-1.15	N/A	N/A	N/A	N/A	-115	+133	N/A

Nomenclature: See separate page